

Loop Breaker

User Manual

VST3 Audio Plugin

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1. INTRODUCTION

Welcome to Loop Breaker, a creative loop-mangling VST3 plugin from Glow Machine, LLC. Loop Breaker combines an MPC-style sampler pad bank with a game-like modifier system that automatically transforms your loops in musically interesting ways.

Load up to 8 audio loops onto the pad grid, start your DAW's transport, and watch as the modifier engine randomly applies effects, slices, pitch shifts, speed changes, and more to your selected pads. The result is an ever-evolving, generative remix of your source material that is perfect for live performance, sound design, or creative exploration.

To get the most out of Loop Breaker, we recommend reading this manual in its entirety. However, if you're eager to dive in, jump straight to the Quick Start section.

Key Features

- 8 stereo audio buffer pads with waveform display and loop progress indicators
- Automatic modifier system that randomly transforms loops at configurable bar intervals
- 24 modifier types spanning buffer transforms, channel effects, master effects, and special operations
- Per-modifier probability sliders with DAW automation and MIDI CC control
- Per-pad target probability to weight which pads receive modifiers
- 4 modifier preset snapshot slots (A-D) for instant recall of complex states
- Multi-output routing: 8 independent stereo output buses plus a master mix bus
- Full MIDI control: pad selection, modifier toggle, and preset recall via MIDI notes
- Drag-and-drop sample loading from your OS file browser
- Parts system: divide loops into 1-4 equal sections for structured arrangements
- Multiple visual themes
- Complete DAW state save and restore

2. SYSTEM REQUIREMENTS

Supported Formats

Loop Breaker is distributed as a VST3 plugin.

Platform

- macOS 10.13 or later (Intel or Apple Silicon)
- A DAW that supports VST3 instruments (Ableton Live, Logic Pro via wrapper, Reaper, FL Studio, Bitwig, etc.)

Audio File Formats

Loop Breaker can load the following sample formats via drag-and-drop or the file chooser:

- WAV (.wav)
- AIFF (.aif, .aiff)
- FLAC (.flac)
- MP3 (.mp3)

Important: DAW Buffer Size

NOTE: Loop Breaker performs real-time time-stretching and pitch-shifting on the audio thread. These DSP operations require large, contiguous blocks of samples. Set your DAW's audio buffer size to the LARGEST available value (typically 2048 or 4096 samples). Using a small buffer (e.g. 64 or 128 samples) will cause audible glitching, clicking, and tearing when speed or pitch modifiers are active. In most DAWs this setting is found under: Preferences > Audio > Buffer Size (or Block Size / Device Buffer).

3. INSTALLATION

macOS

Run the Loop Breaker .pkg installer. The installer places the VST3 bundle into the standard system plugin folder:

```
/Library/Audio/Plug-Ins/VST3/Loop Breaker.vst3
```

After installation, relaunch your DAW and scan for new plugins. Loop Breaker will appear in the Instrument category under "Glow Machine Audio".

Uninstallation

To remove Loop Breaker, run the included uninstall script or manually delete the VST3 bundle from the above path.

4. QUICK START

Follow these steps to get up and running quickly:

Step 1: Load Samples

Drag WAV, AIFF, FLAC, or MP3 files from Finder onto any pad in the 2x4 grid. Alternatively, click an empty pad to open a file chooser dialog.

Step 2: Start the DAW Transport

Loop Breaker follows the host timeline. Press Play in your DAW and all loaded buffers will begin looping. Each pad displays a waveform with a moving playhead.

Step 3: Select Target Pads

Click pads to toggle their selection on or off. Selected pads glow and will be targeted by the next modifier. If no pads are selected when the modifier timer fires, 1-4 pads are chosen automatically.

Step 4: Watch the Modifier Display

The upcoming modifier name and variant details appear in the Session tab header. A progress bar and countdown show when the next modifier will fire.

Step 5: Adjust Probabilities

Switch to the Probability tab to control how likely each modifier type is to appear. Set a slider to 0 to disable that modifier entirely.

TIP: Use the Master Volume knob in the Session tab to compensate if the default per-pad headroom reduction (-12 dB) makes the mix too quiet.

5. USER INTERFACE OVERVIEW

The Loop Breaker plugin window is organized into a tabbed layout with four main tabs:

Session

The primary performance view. Contains the pad grid, upcoming modifier display, modifier toggle, master volume knob, and preset buttons (A-D).

Probability

Per-modifier probability sliders grouped by category (Buffer, Channel Effect, Master Effect, Special). Also includes per-pad target probability sliders. All sliders are exposed as DAW automation parameters.

Settings

Appearance theme selection, Parts configuration, and Bars-per-Modifier timing control.

Help

Built-in reference documentation for all plugin features, controls, and modifier types.

6. SESSION TAB

The Session tab is the main performance view and the default tab when the plugin opens.

Pad Grid

The 2x4 pad grid represents the 8 audio buffers. Each pad displays the loaded sample's filename, a waveform thumbnail, and a playhead indicator showing the current loop position. Pads change color to indicate their state:

- Empty: dark/unlit - no sample loaded
- Loaded: standard color - sample loaded, not selected
- Selected: highlighted/glowing - will be targeted by next modifier
- Playing: animated indicator when the transport is running

Pad Interactions

Click	Toggle pad selection on / off
Right-click	Open context menu (Load Sample, Remove Sample, MIDI Learn, Clear MIDI Note)
Drag & drop file onto pad	Load audio file into that pad
Shift + Click	Enter MIDI learn mode for this pad; play a MIDI note to assign it; click again to cancel
Cmd + Click (macOS)	Clear the MIDI note assignment for this pad
Alt + Click	Clear the MIDI note assignment for this pad (alias)
Shift + Cmd + Click (macOS)	Remove the loaded sample from this pad

Upcoming Modifier Display

Located in the session header area, this display shows the name and variant details of the next modifier that will be applied. A progress bar and countdown (in bars/seconds) indicate how much time remains before the modifier fires. When the modifier triggers, the targeted pads flash to provide visual feedback.

Modifiers Toggle

This button enables or disables the modifier scheduling system entirely. When disabled, no modifiers will be applied and the countdown pauses. The toggle supports MIDI note assignment via right-click context menu or Shift+Click (MIDI Learn) / Cmd+Click (clear).

Master Volume Knob

A rotary knob that controls the overall output level from -12 dB to +12 dB. This gain is applied equally to all pads and all output buses. The parameter is exposed for DAW automation.

Modifier Preset Buttons (A-D)

Four preset slots allow you to snapshot and recall the complete modifier state across all 8 pads. This captures buffer transform parameters (speed, stretch, pitch, slicing, ping-pong) and all channel effects settings (reverb, delay, filter, tremolo, chorus, auto-pan, volume ramp).

Click (empty slot)	Save the current modifier states into this preset
Click (filled slot)	Recall the saved modifier states from this preset
Double-click	Force-save (overwrite) the current states into this preset
Right-click	Open context menu: Save, Recall, Clear, MIDI Learn, Clear MIDI
Shift + Click	Enter MIDI learn mode for this preset slot
Cmd + Click (macOS)	Clear the MIDI note assignment for this preset

NOTE: Presets do NOT save or restore Parts settings, pad file assignments, probability weights, or timing configuration. Presets are saved with your DAW session and restored on reload.

7. PROBABILITY TAB

The Probability tab gives you fine-grained control over which modifiers appear and which pads are targeted.

Modifier Probability Sliders

Each modifier type has a probability slider ranging from 0.0 to 1.0. Set a slider to 0 to completely disable that modifier. All enabled sliders are normalized automatically: a modifier set to 2.0 is twice as likely to appear as one set to 1.0.

Sliders are grouped by category: Buffer, Channel Effect, Master Effect, and Special. Each slider is exposed as a DAW automation parameter.

MIDI CC Control

Each probability slider can be mapped to a MIDI CC. Click the CC button beside a slider to enter MIDI CC learn mode, then move any CC knob or fader on your controller. The assignment is saved with the DAW session. This allows you to drive modifier probabilities from external hardware or automation lanes.

Pad Target Probability

At the bottom of the Probability tab, a section of 8 sliders (Pad 1 through Pad 8) controls the likelihood that each pad will be automatically selected as a modifier target. A value of 1.0 means the pad is always eligible; 0.0 means it will never be auto-selected. These sliders also support DAW automation and MIDI CC learn.

Reset All to 100%

A button at the bottom of the panel resets all probability sliders to their default value of 1.0 (100%).

8. SETTINGS TAB

Theme

Choose from 11 built-in visual themes. The theme change crossfades smoothly over 500ms. Available themes:

- Arctic Sky
- Daylight
- Gruvbox
- Ivory
- Neon Rave (Dark)
- Pixel Grid
- Silver
- Studio Clean
- Ultraviolet
- Vintage Ember
- Warm Paper

Parts

Split each buffer into 1 to 4 equal sections (labeled A through D). This allows you to work with structured musical arrangements. The Part change takes effect on the next modifier trigger when the transport is running. The "Switch Part" modifier type will automatically move between parts when enabled.

Bars / Modifier

This slider (range: 1 to 16 bars) controls how many bars elapse between each modifier application. At 1 bar, modifiers fire rapidly for chaotic, fast-evolving textures. At 16 bars, changes are slow and gradual. The default is 4 bars.

9. MODIFIERS REFERENCE

Loop Breaker's modifier system is at the heart of its creative engine. Every N bars (configured in the Settings tab), a randomly chosen modifier is applied to the user-selected pads. If no pads are selected, 1-4 pads are chosen automatically.

Modifiers fall into four categories:

Buffer Transforms

These modifiers directly alter how the audio buffer is played back.

Reverse

Flips the playback direction of the buffer. If the buffer is already playing in reverse, it returns to forward playback.

Speed

Changes the playback rate to one of four discrete values: 0.25x, 0.5x, 1.0x, or 2.0x. Time-stretching is used to maintain the original pitch.

Stretch

A time-stretching variant that changes tempo without affecting pitch. Available ratios include 0.25x, 0.5x, 1.0x, and 2.0x.

Pitch Up Octave

Raises the pitch of the buffer by one octave (+12 semitones).

Pitch Down Octave

Lowers the pitch of the buffer by one octave (-12 semitones).

Beat Slice

Subdivides the buffer into note-length slices (1/4, 1/8, 1/8T, 1/16, 1/32, 1/64) and plays them in a randomized order, creating glitchy, rearranged patterns.

Arp Slice

Divides the buffer into a configurable number of slices and plays a short repeating arpeggio-like sequence from those slices for a set number of bars.

Slice Repeater

Selects a single slice from the buffer and stutters/repeats it a configurable number of times (4, 8, 16, or 32 repetitions).

Ping Pong

Alternates playback direction on each loop cycle at a musical note division (whole note, half note, quarter note, eighth note, or sixteenth note).

Channel Effects

These modifiers enable or adjust effects on individual buffer channel strips.

Delay

Enables the delay effect on the buffer's channel strip. Variants include different delay time divisions (1/4, 1/8, 1/8D, 1/8T, 1/16), wet mix levels, feedback amounts, optional ping-pong mode, and wow/flutter modulation.

Delay Dub Burst

A special delay variant that creates an intense burst of delay feedback for a short duration, then fades back.

Reverb

Enables reverb on the channel strip with a configurable wet/dry mix (25%, 50%, 75%, or 100%) and a fade-in duration (instant, 1 bar, 2 bars, etc.).

Low-Pass Filter

Enables a low-pass filter that gradually closes over the configured duration, darkening the sound.

High-Pass Filter

Enables a high-pass filter that gradually opens over the configured duration, thinning the low end.

Volume Ramp Down

Gradually fades the buffer volume down over a number of bars.

Tremolo

Applies volume modulation (LFO) to the buffer, creating a rhythmic pulsing effect.

Chorus

Enables a chorus effect with configurable depth, rate, and wet/dry mix for a thicker, wider sound.

Auto-Pan

Modulates the stereo panning of the buffer with a configurable LFO rate and depth, creating movement in the stereo field.

Ducking

Enables sidechain-style ducking that reduces the buffer volume when signal is present, creating a pumping effect.

Master Effects

These modifiers affect the master output and are applied regardless of pad selection.

Master High-Pass

Applies a high-pass filter to the master output, thinning the overall mix. Can ramp up then back down, or jump immediately and ramp back.

Master Low-Pass

Applies a low-pass filter to the master output, darkening the overall mix. Behaves similarly to the Master High-Pass with configurable ramp behavior.

Special Modifiers

Switch Part

Moves all targeted buffers to a different Part (A-D). Only available when Parts is set to more than 1 in the Settings tab.

Quarter-Note Burst

Triggers rapid-fire modifier applications at quarter-note intervals for a configurable number of bars (1, 2, or 4), creating intense bursts of change.

Reset All

Removes all active modifiers, turns off all effects, and returns playback speed, pitch, and direction to their default state. This is your "panic button."

10. MIDI CONTROL

Loop Breaker accepts MIDI note input for pad selection, modifier toggling, and preset recall. MIDI CC input is used for probability slider control.

Default MIDI Note Map

By default, pads are mapped to MIDI notes matching the General MIDI drum layout:

Note #	Name	Pad	Position
36	C1	Pad 1	Bottom-left
37	C#1	Pad 2	Bottom row
38	D1	Pad 3	Bottom row
39	D#1	Pad 4	Bottom-right
40	E1	Pad 5	Top-left
41	F1	Pad 6	Top row
42	F#1	Pad 7	Top row
43	G1	Pad 8	Top-right

Note behavior: Note-on toggles the pad selection. Note-off is ignored. Velocity is ignored.

MIDI Learn

To reassign a pad to a different MIDI note: Shift+Click the pad (or right-click and choose "MIDI Learn"), then play the desired note on your controller. The new mapping is saved with the DAW session.

To clear a MIDI note assignment: Cmd+Click (macOS) or Alt+Click, or right-click and choose "Clear MIDI Note".

Modifier Toggle via MIDI

The Modifiers on/off toggle button can also be assigned to a MIDI note. Right-click the toggle and choose MIDI Learn, or Shift+Click it. Use Cmd+Click to clear the assignment.

Preset Recall via MIDI

Each preset slot (A-D) can be assigned to a MIDI note for instant recall from a hardware controller. Use the right-click context menu or Shift+Click on a preset button to enter MIDI Learn mode.

MIDI CC for Probabilities

In the Probability tab, click the CC button next to any slider to enter MIDI CC learn mode. Move

a knob or fader on your controller to assign it. This lets you drive modifier probabilities with hardware faders, LFOs, or DAW MIDI automation.

TIP: Loop Breaker is compatible with most drum pad controllers (Akai MPD, NI Maschine, Novation Launchpad, etc.) and any standard MIDI keyboard.

11. MULTI-OUTPUT ROUTING

Loop Breaker exposes 8 independent stereo output buses (plus a master mix bus) for a total of up to 18 output channels.

Output Bus Layout

Bus	Channels	Source
Master Mix	1-2	Sum of all pads
Output 1	3-4	Pad 1
Output 2	5-6	Pad 2
Output 3	7-8	Pad 3
Output 4	9-10	Pad 4
Output 5	11-12	Pad 5
Output 6	13-14	Pad 6
Output 7	15-16	Pad 7
Output 8	17-18	Pad 8

To use multi-output routing, enable the additional output buses on the plugin instance in your DAW. Each pad's audio is then routed to its own mixer channel, allowing independent processing, effects, and mixing within the DAW.

If the DAW only activates a subset of buses, or does not support multi-output, the remaining buffers are automatically folded into the master mix output.

12. DAW AUTOMATION

Loop Breaker exposes the following parameters for DAW automation:

Automatable Parameters

- Master Volume (-12 dB to +12 dB)
- Per-modifier probability sliders (one for each of the 24 modifier types)
- Per-pad target probability sliders (Pad 1 through Pad 8)

These parameters appear in your DAW's automation lane selector under the Loop Breaker plugin. You can draw automation curves, use LFOs, or link them to MIDI CC via the plugin's built-in learn feature.

State Save & Restore

All plugin state is saved and restored with the DAW session, including:

- Loaded sample file paths (samples are reloaded from disk on session open)
- Pad MIDI note assignments
- Modifier toggle MIDI note
- Preset MIDI note assignments
- All probability slider positions and MIDI CC mappings
- Pad target probability positions and MIDI CC mappings
- Modifier preset snapshots (A-D)
- Parts configuration (number of parts, active part)
- Bars-between-modifiers setting
- Theme selection
- Modifiers enabled/disabled state

13. SAMPLE LOADING

Drag & Drop

Drag one or more audio files from Finder directly onto the pad grid. If you drag multiple files at once, they will be loaded sequentially starting from the pad you drop onto.

File Chooser

Click an empty pad to open a file chooser dialog. Navigate to your sample and click Open. You can also right-click any pad and select "Load Sample" from the context menu.

Removing a Sample

Right-click the pad and choose "Remove Sample", or use Shift+Cmd+Click (macOS).

NOTE: Sample loading is performed off the audio thread. You can load new files at any time without interrupting playback. If a loaded file goes missing (e.g. sample drive unmounted), the pad shows an error state and outputs silence.

14. PARTS SYSTEM

The Parts system allows you to divide each audio buffer into 1 to 4 equal-length sections, labeled A through D. This is useful when your loops have distinct musical sections (e.g. verse, chorus, bridge, outro) that you want to navigate between.

Configuring Parts

In the Settings tab, use the Parts dropdown to select 1 part (default), 2 parts, 3 parts, or 4 parts. The change takes effect on the next modifier trigger when the transport is running.

Switching Parts

The "Switch Part" modifier (visible in the Probability tab under the Special category) will automatically move buffers to a different part when it fires. This modifier is only available when Parts is set to more than 1.

15. THEMES

Loop Breaker includes 11 built-in visual themes that change the entire color palette of the plugin interface. Themes crossfade smoothly over 500 milliseconds when switched.

Available Themes

Neon Rave (Dark)

The default theme. Dark background with vibrant neon accent colors. Ideal for dimly-lit studios.

Arctic Sky

Cool blue tones with a clean, icy aesthetic.

Daylight

Bright, light theme with warm neutral tones for well-lit environments.

Gruvbox

Inspired by the popular Gruvbox color scheme. Warm, retro earth tones.

Ivory

Minimal, elegant off-white palette.

Pixel Grid

Retro digital aesthetic with pixel-art-inspired colors.

Silver

Sleek metallic gray palette.

Studio Clean

Professional, neutral tones designed for mixing sessions.

Ultraviolet

Deep purples and electric blues for a futuristic vibe.

Vintage Ember

Warm amber and brown tones with a vintage analog feel.

Warm Paper

Soft, paper-like tones for a gentle reading experience.

16. TIPS & KNOWN ISSUES

Tips

- Start with the Bars / Modifier slider at 4 bars, then experiment with shorter intervals for more chaos or longer intervals for subtle evolution.
- Use the Reset All modifier (ensure its probability is non-zero) as a natural recovery mechanism that periodically returns your loops to their original state.
- Assign modifier probability sliders to MIDI CC on your controller to perform live probability adjustments during a set.
- Use multi-output routing to apply DAW-side effects (like EQ or compression) to individual pads independently.
- Save modifier presets (A-D) for different "scenes" during a performance and trigger them via MIDI for instant transitions.
- Lower the probability of extreme modifiers (Pitch Up/Down, Beat Slice) if you want more subtle, musical evolution.

Known Issues

- Modifier display and scheduling are only active while the plugin editor window is open. Closing the window suspends modifiers; re-opening resumes them.
- Octave pitch-shifting may produce audible artifacts at extreme settings. Staying within +/-1 octave typically sounds cleanest.
- Small DAW buffer sizes (< 512 samples) may cause tearing or glitching with time-stretch effects. Use 2048+ for best results.

17. TECHNICAL INFORMATION

Audio Engine

Loop Breaker is built with the JUCE framework. The audio engine features:

- Linear interpolation for high-quality sample rate conversion
- Smoothed parameter changes to prevent audio artifacts
- Atomic variables for thread-safe real-time parameter updates
- Crossfaded looping for seamless audio transitions
- Real-time time-stretching via the SoundTouch library
- Per-channel DSP: reverb, delay (with ping-pong and wow/flutter), low-pass filter, high-pass filter, tremolo, chorus, auto-pan, ducking, and limiter
- Off-thread sample loading to avoid audio thread interruption

Plugin Format

Property	Value
Format	VST3
Category	Instrument (Synth)
MIDI Input	Yes
MIDI Output	No
Manufacturer	Glow Machine Audio
Plugin Code	TsPI
Manufacturer Code	GloM
AU Export Prefix	LoopBreakerAU
AAX Identifier	com.glowmachineaudio.LoopBreaker

18. CREDITS & CONTACT INFORMATION

Credits

Loop Breaker is developed by Glow Machine, LLC.

Third-party libraries:

- JUCE Framework - JUCE is copyright Raw Material Software Limited.
- SoundTouch Audio Processing Library - used for real-time time-stretching and pitch-shifting. SoundTouch is copyright Olli Parviainen.

Contact

Glow Machine, LLC

Website: www.glowmachineaudio.com

For support, bug reports, and feature requests, please visit our website.

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